

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

B O A 2 0 4 - R

Hands-on with multi-Region architectures

Seth Eliot

Principal Developer Advocate
AWS Developer Relations

Gunnar Grosch

Principal Developer Advocate
AWS Developer Relations



© 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Agenda: Hands-on with multi-Region architectures

1. Let's talk

Why multi-Region:
Slides

2. Let's build

Building multi-Region:
Demo



Why multi-Region?

AWS Regions and Availability Zones (AZs)

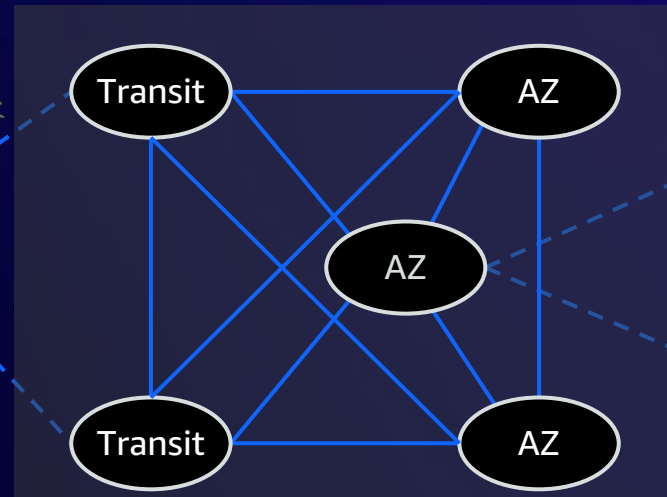
AWS Regions are physical locations around the world where we cluster data centers

32 AWS Regions worldwide



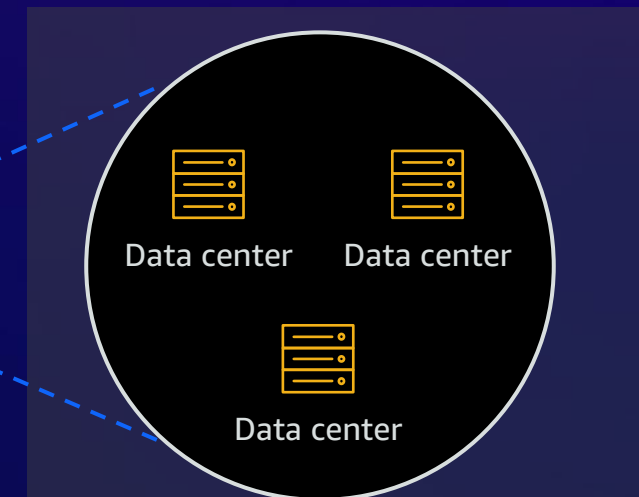
- AWS Regions
- Announced Regions

Each AWS Region has multiple AZs



A Region is a physical location in the world

Each AZ includes one or more discrete data centers



Data centers, each with redundant power, networking, and connectivity, housed in separate facilities

When might you need multi-Region architecture?



Resilience



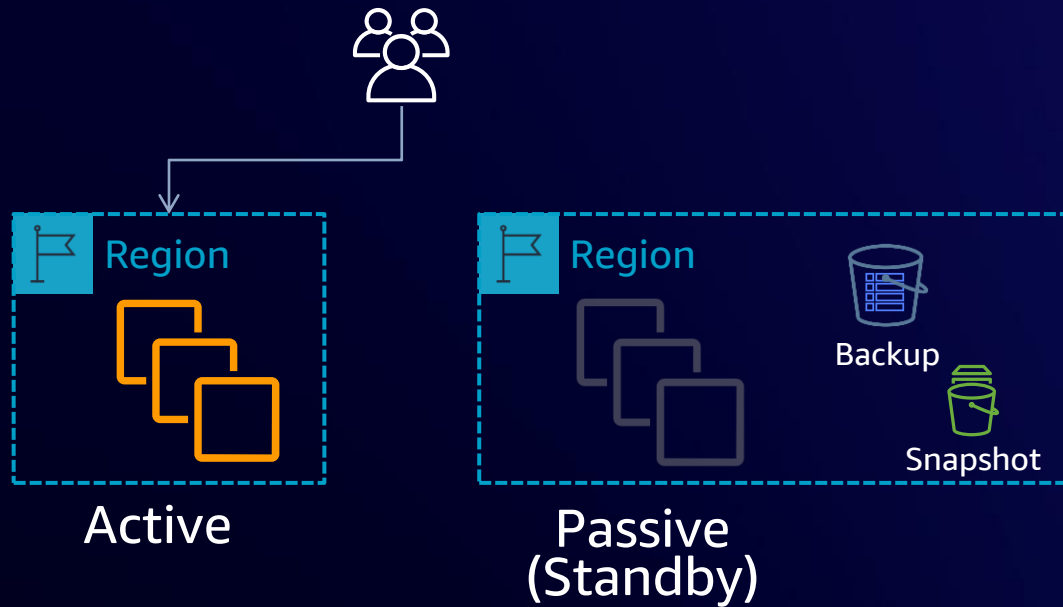
Regulatory
requirements



Global
performance

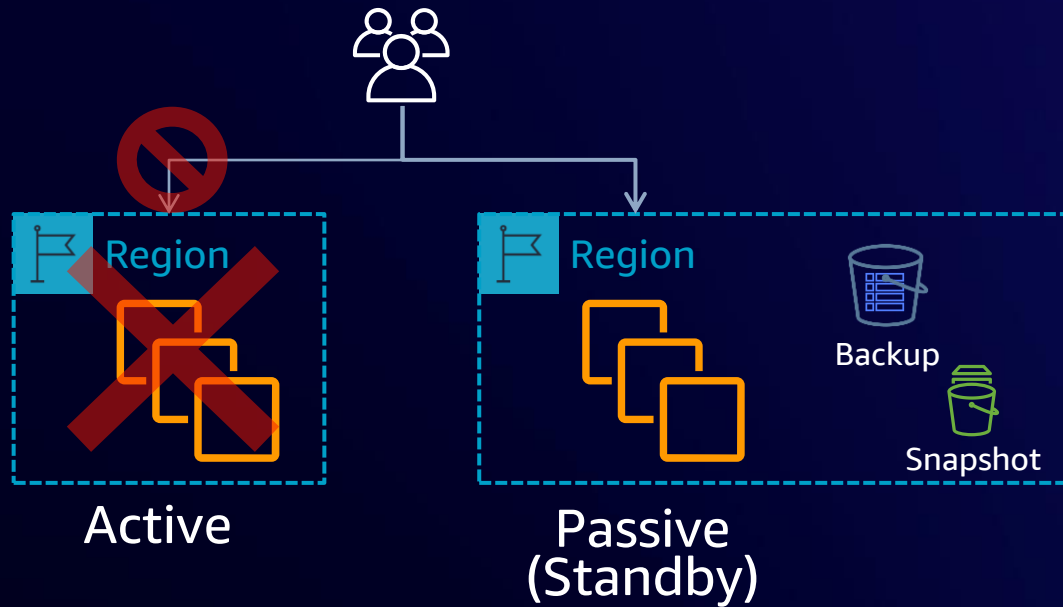
Two different approaches to multi-Region

Active-passive and active-active



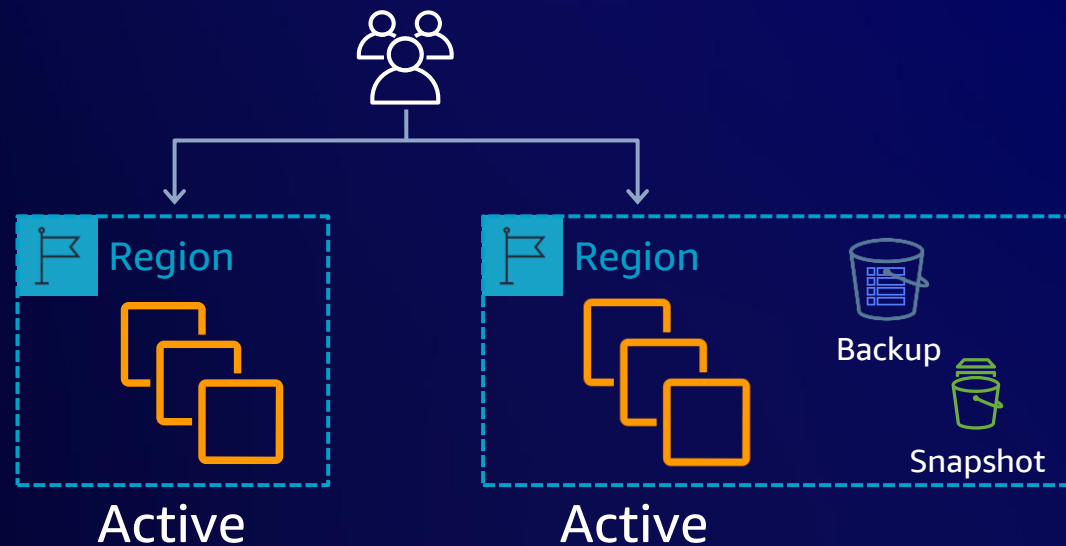
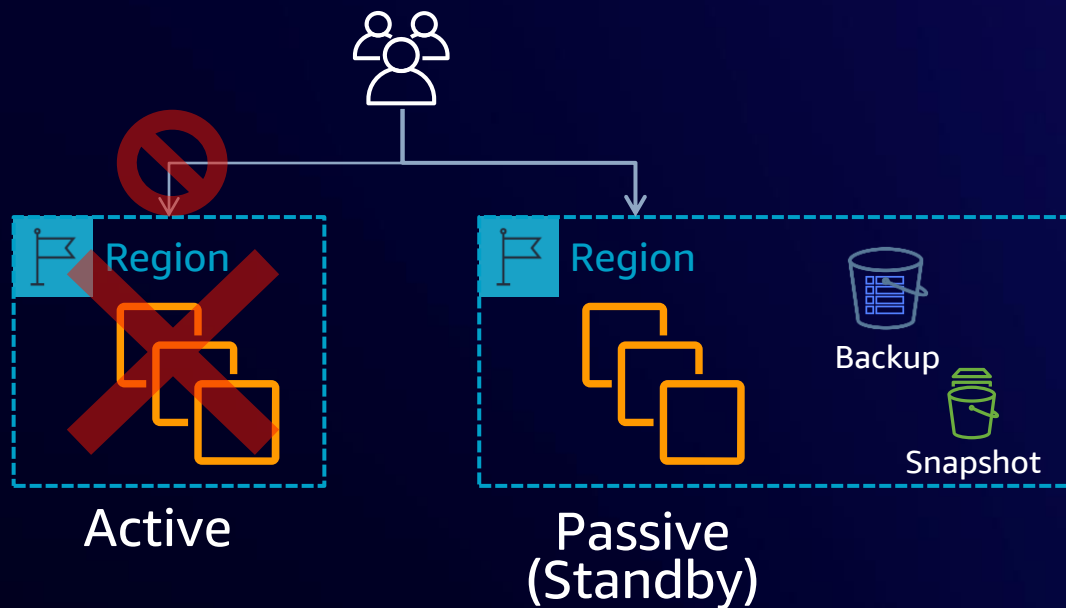
Two different approaches to multi-Region

Active-passive and active-active



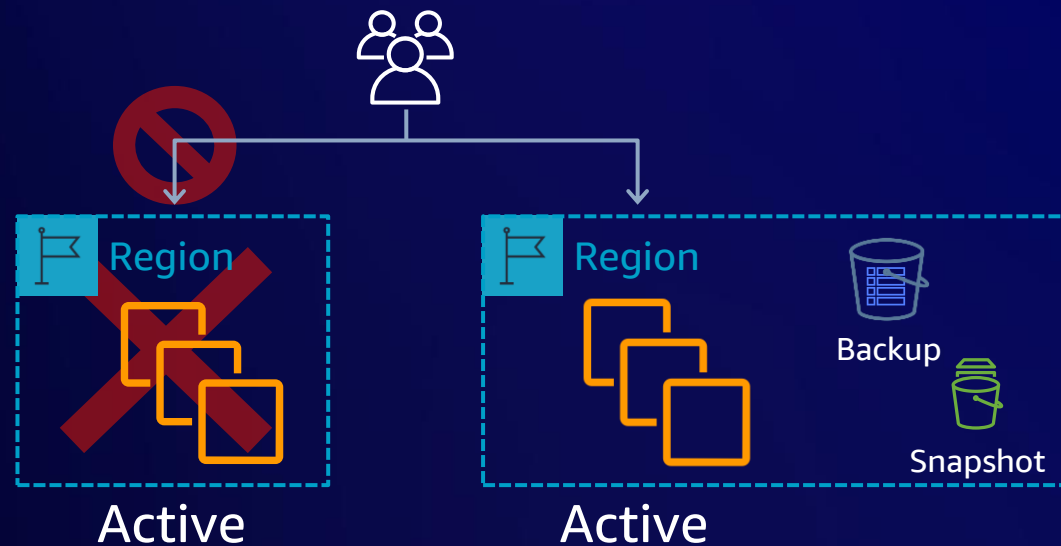
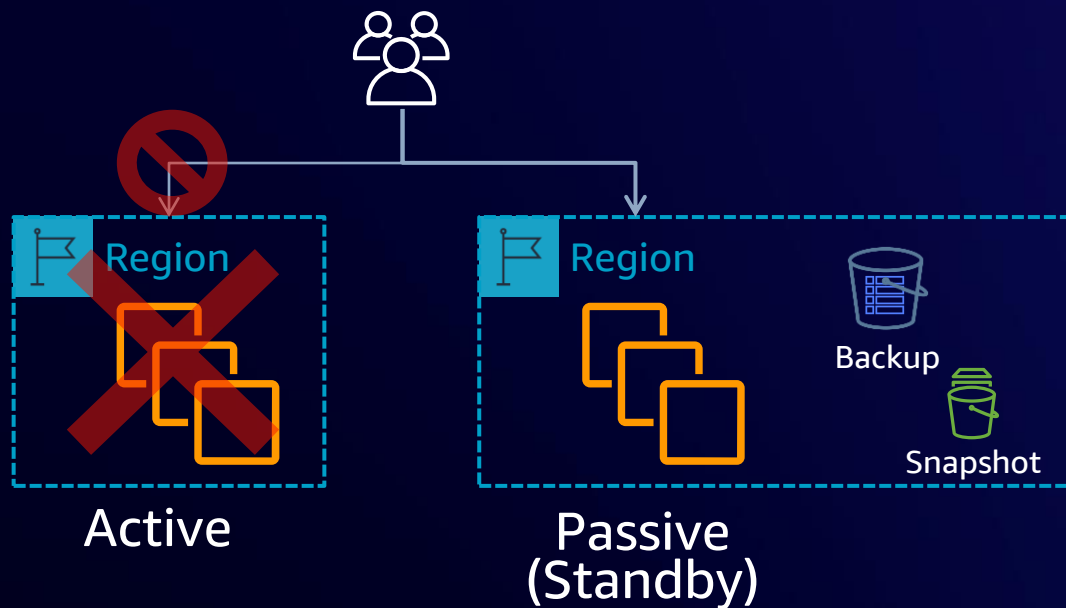
Two different approaches to multi-Region

Active-passive and active-active



Two different approaches to multi-Region

Active-passive and active-active



When do you need a multi-Region active-passive strategy?



Disaster recovery (DR)

If you need to mitigate the risk of an event spanning multiple data centers a significant distance away from each other



Regulatory requirements

If you need to store assets more than 60 miles/
100 kilometers apart

When do you need a multi-Region active-active strategy?



Extreme HA
and also DR

Greater than 99.99%
availability through Regional
redundancy



Regulatory
requirements

Data must be constrained
to user locale



Global
performance

If you need service
endpoints close to users or
separate stacks per location

Regulatory requirements and global performance



Regulatory
requirements



Global
performance



Regulatory requirements:

- Data stays local to users
- Data backed up far away

Better performance:

- Users routed to local stack

Backups to another Region

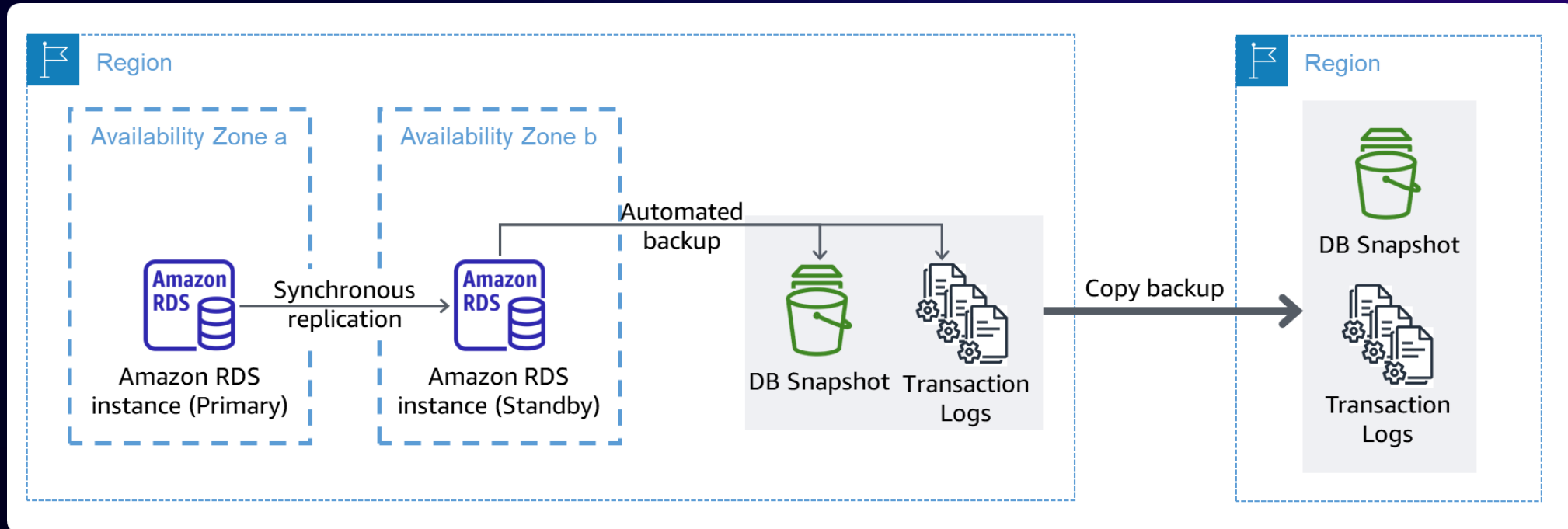
Almost always a good idea



Disaster
recovery (DR)



Regulatory
requirements



Extra layer of protection

The good: Simple to implement – AWS Backup

The bad: Extra cost

The recovery: Can be time-consuming

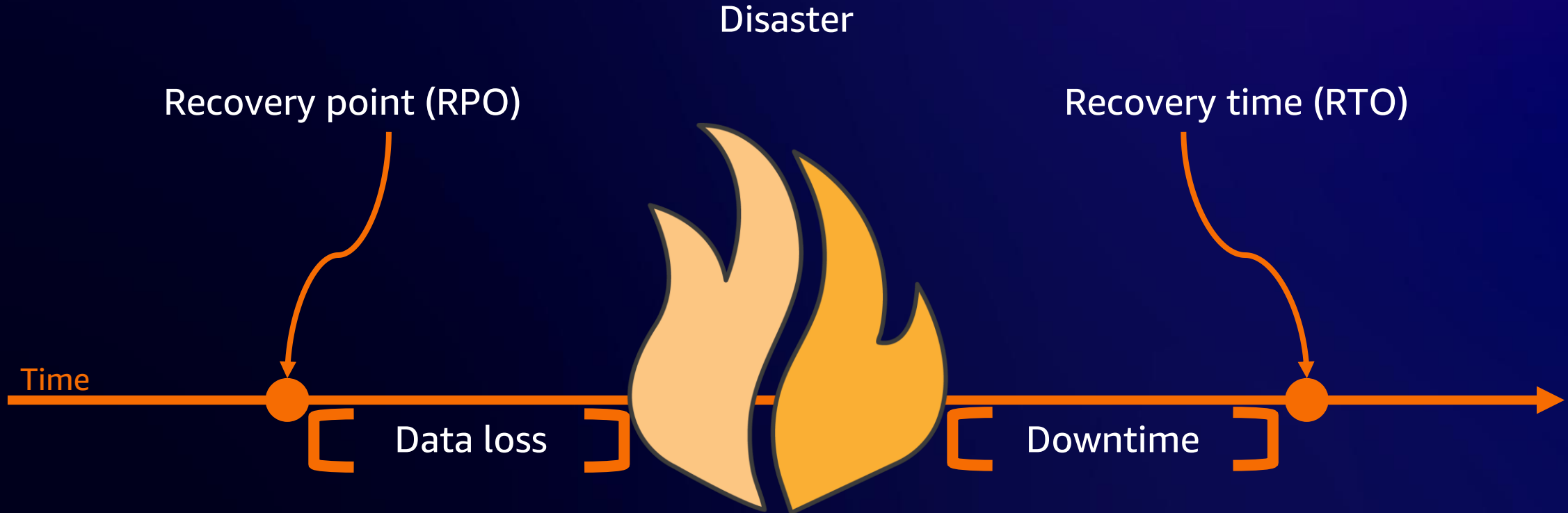
Disaster recovery objectives



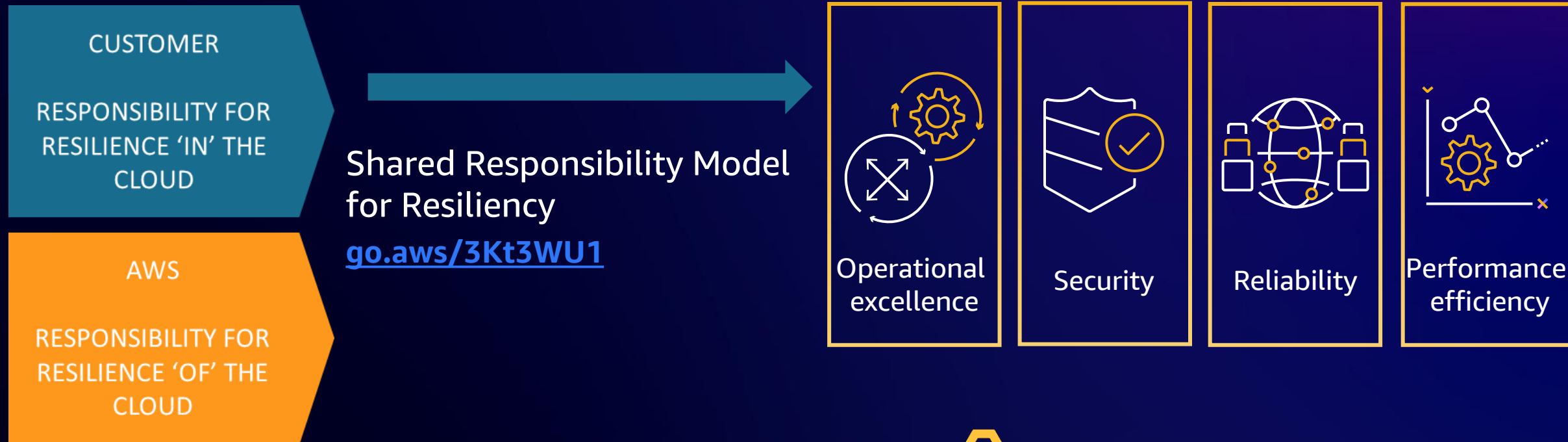
Disaster
recovery (DR)

How much data can you afford
to re-create or lose?

How quickly must you recover?
What is the cost of downtime?



Your responsibility: Use best practices



AWS Well-Architected

aws.com/well-architected

AWS Resilience Hub: Key capabilities

Assess



Assess whether or not
RTO/RPO is met

Run daily or with CI/CD

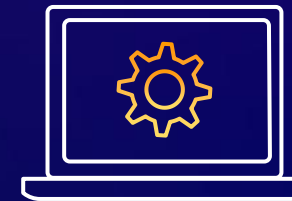
Resilience recommendations



Best practices to
optimize for

- Best RTO/RPO
- Cost
- Minimal changes

Operational recommendations



Alarms

Standard operating
procedures (SOP)

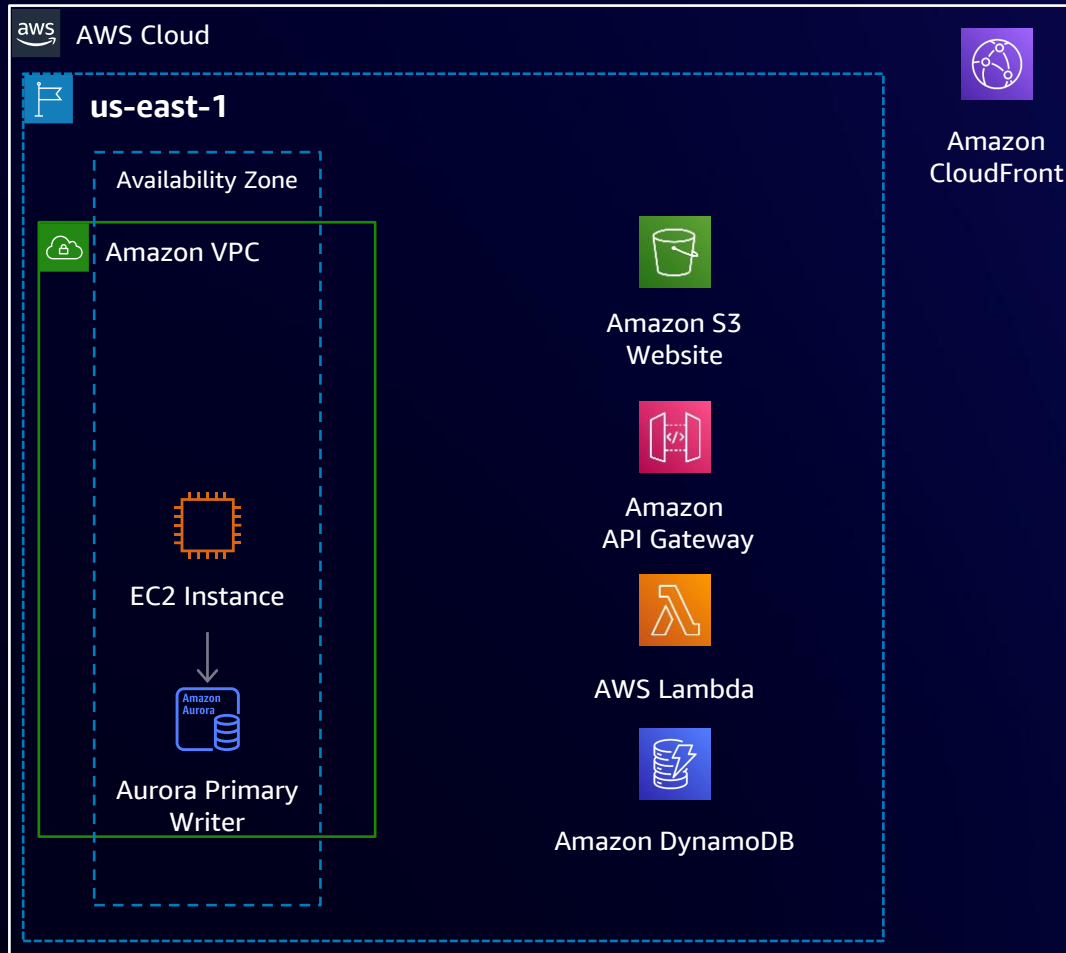
Fault injection experiments

Let's build



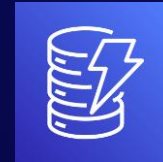
Demo architecture

And some AWS service features for multi-Region



Amazon S3

S3 Multi-Region Access Points (MRAP)
S3 Cross-Region Replication (CRR)



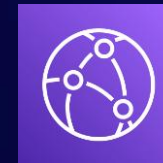
Amazon DynamoDB

DynamoDB global tables



Amazon Aurora

Aurora global database



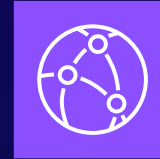
Amazon CloudFront

CloudFront origin failover

AWS services used to route traffic among Regions



Amazon Route 53



Amazon CloudFront



AWS Global Accelerator

What is it

DNS service

CDN

Networking service

Automatic failover

using data plane

Failover routing policy

Origin failover

Endpoint health checks

Manual failover:

using data plane

R53 ARC; STOP

-

-

using control plane

Change weights

Update distribution

Dial traffic

Learn more

ARC308 | Best practices for creating multi-Region architectures on AWS | Breakout
ARC401 | Multi-Region or single-Region? Considerations and architectures | Chalk Talk
FSI301 | Architecting critical multi-Region, active/active financial systems | Chalk Talk

[Workshop] Plan for Disaster Recovery (DR) – Hot Standby
https://bit.ly/dr_hot_standby

[Whitepaper] “Disaster Recovery of Workloads on AWS: Recovery in the Cloud”
bit.ly/DR_AWS

[Blog posts] “Disaster Recovery” series
bit.ly/aws-dr-blog

[Blog posts] “Creating a Multi-Region Application with AWS Services” series
go.aws/3EJO13A

“What is AWS Resilience Hub?”
docs.aws.amazon.com/resilience-hub/latest/userguide/what-is.html



Thank you!



Please complete the session survey in the mobile app

Seth Eliot

✉ seliot@amazon.com
[in linkedin.com/in/setheliot](https://www.linkedin.com/in/setheliot)
[X @setheliot](https://twitter.com/setheliot)

Gunnar Grosch

✉ ggrosch@amazon.com
[in linkedin.com/in/gunnargrosch](https://www.linkedin.com/in/gunnargrosch)
[X @gunnargrosch](https://twitter.com/gunnargrosch)